

BIOMECHANICS (EPHE 341)

Angular Kinetics 1

Torque and moment of force — how a push becomes a turn

By the end of this lecture you should be able to...

- **Define** torque as force times the perpendicular distance from its line of action to an axis
- **Assign** a sign with the right-hand rule, and add torques that act about the same axis
- **Distinguish** an external moment from the internal moment that opposes it
- **Account** for lateral, anterior and posterior trunk bending as ways of moving a line of action
- **Find** a moment arm by either route — resolve the distance or resolve the force — and get the same answer
- **Explain** why the same muscle force produces a different joint torque at a different joint angle
- **Read** a joint moment graph through the gait cycle and name the muscle group responsible

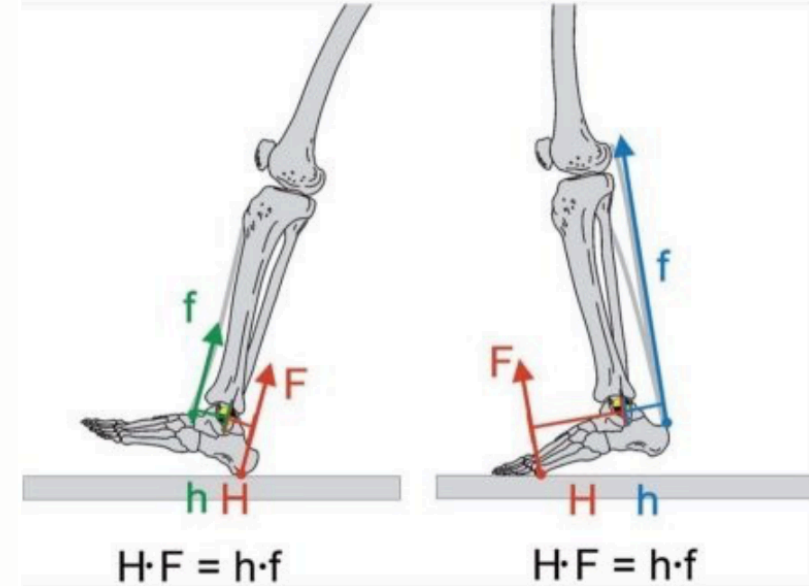
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A force does not only push things along

We have been treating a force as something that causes **translation**. But a force can be applied almost anywhere on a rigid object, and when it is, it also tends to cause **rotation**.

Torque is the tendency of a force to rotate an object about an axis.

- Also called the **moment of force**, or simply the **moment**.
- Think of it as a rotary force — the turning effect of a push.
- A torque can be present *even when nothing moves*. Standing still on one leg, the ground is applying a large torque about your hip, and your abductors are applying an equal and opposite one.



The foot is a lever. The ground pushes up with F at a distance H from the ankle; the Achilles pulls with f at a much shorter distance h . Nothing is rotating, and yet $H \cdot F = h \cdot f$.

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The definition

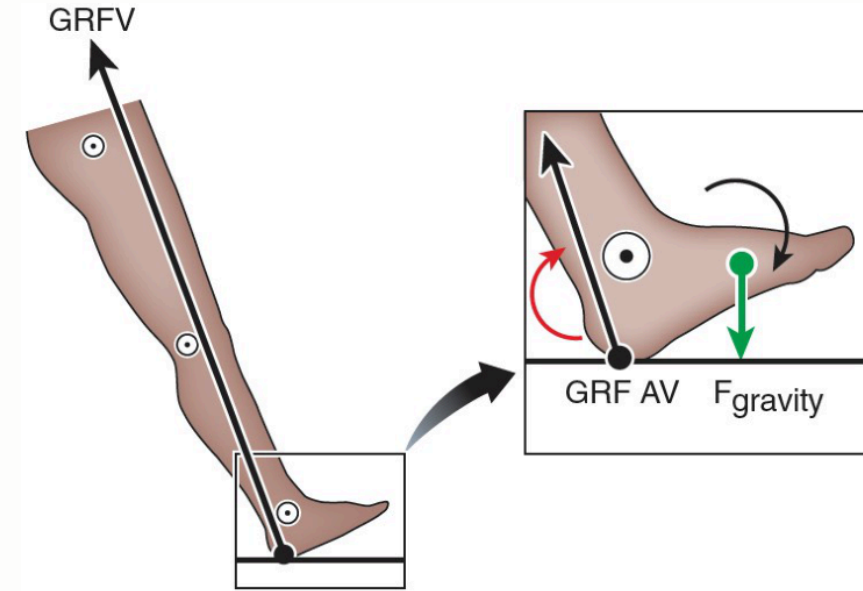
$$T = F \cdot d_{\perp}$$

$$M = F \cdot d_{\perp}$$

Torque is the result of a force **F** applied a certain perpendicular distance **d_⊥** from an axis of rotation. T and M are the same quantity; both letters are in common use.

Units. Newtons times metres: **N·m**. Not joules — a joule is also a newton-metre, but it is a force moved *along* its own line, which is a completely different thing. Keep torque in N·m and never call it a joule.

Torque is influenced by three things, and only three: the **magnitude** of the force, its **line of action**, and its **point of application**. The last two together fix **d_⊥**.



The ground reaction force vector has a magnitude, a line of action and a point of application — the centre of pressure. Change any one and the moment at the ankle changes.

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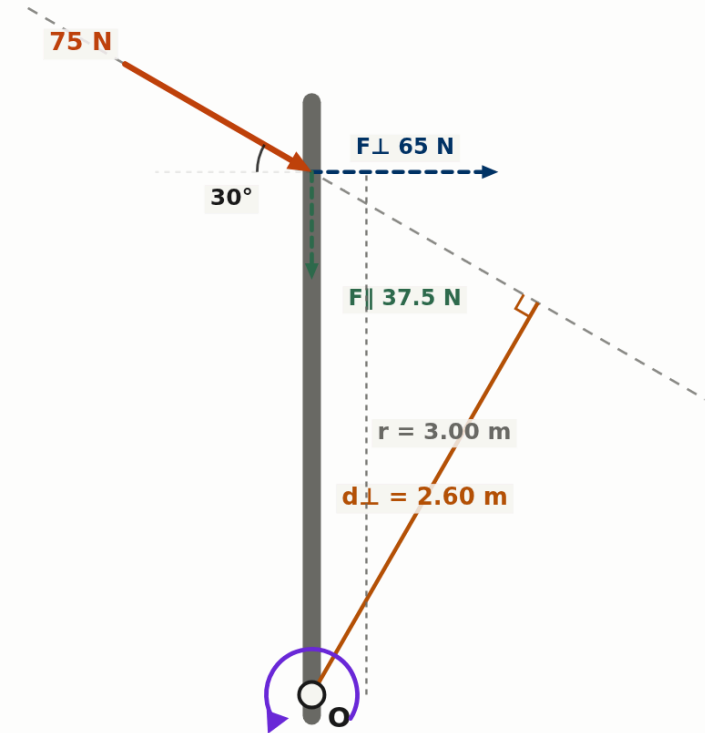
The moment arm

The **moment arm** is the perpendicular distance from the **line of action** of the force to the **axis of rotation**.

- It is measured to the *line*, extended as far as you like in both directions — not to the point where the force happens to be applied.
- A force whose line passes *through* the axis has a moment arm of zero and produces no torque, however large it is.

The axis need not be a hinge. It may be fixed, like the pin of a seesaw or the axle of a door. Or it may be free — a diver in the air rotates about her own centre of gravity, with nothing holding her there at all.

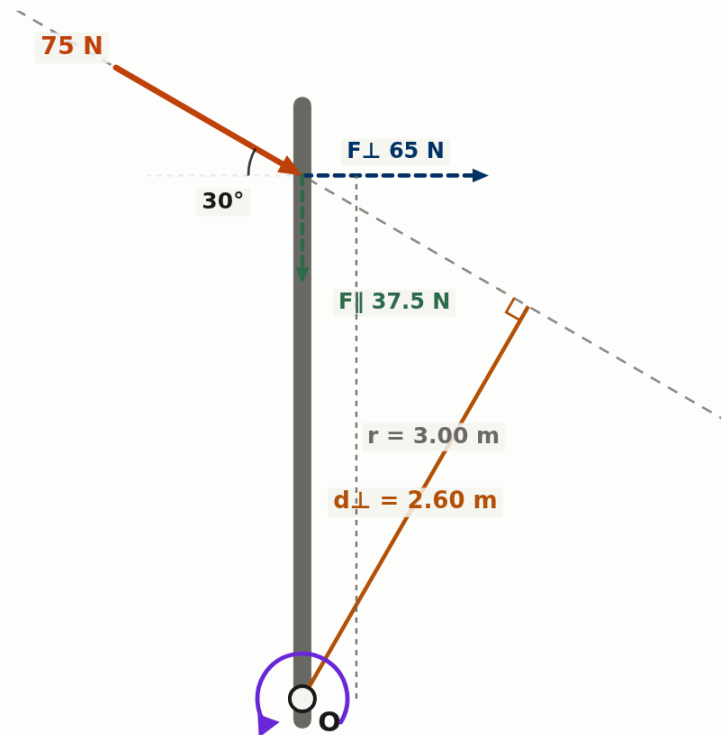
Naming the axis first is not a formality. The same force has a different moment arm, and often a different sign, about two different axes. Half of all wrong answers in this topic are answers to a different question.



the moment arm is the perpendicular from the axis to the **LINE**,
not the distance to where the force happens to be applied

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Two routes to the same number



A — resolve the distance

$$d_{\perp} = r \cos \theta = 3.00 \times 0.866 = 2.598 \text{ m}$$

$$M = F \cdot d_{\perp} = 75 \times 2.598 = 195 \text{ N}\cdot\text{m}$$

B — resolve the force

$$F_{\perp} = F \cos \theta = 75 \times 0.866 = 65.0 \text{ N}$$

$$M = F_{\perp} \cdot r = 65.0 \times 3.00 = 195 \text{ N}\cdot\text{m}$$

$M_o = -195 \text{ N}\cdot\text{m}$

clockwise, into the page

$F_{\parallel} = 37.5 \text{ N}$ runs through O.

No moment arm, no moment — it only compresses.

Angle θ from the perpendicular

Distance r

Force F

As set up, this is his worked example: **75 N** striking a post **3 m** above the axis at **30°** to the perpendicular, giving **195 N·m clockwise** by either route. Push θ to $\pm 80^\circ$ and the force nearly runs along the post, through O: the moment arm all but disappears and a large force achieves almost nothing.

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Torque is a vector

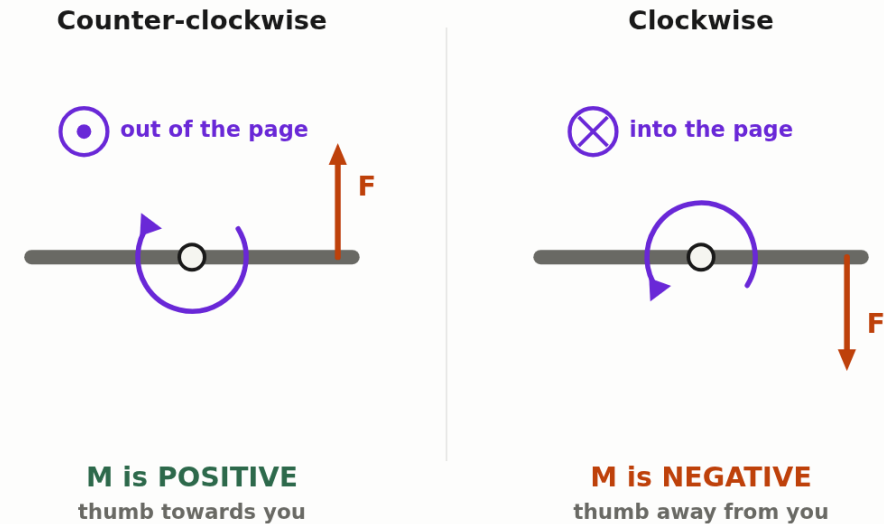
A torque has a magnitude and a direction, so it is a vector. Its direction is along the **axis of rotation**, not along the force.

The right-hand rule. Curl the fingers of your right hand in the direction the force tends to turn the object. Your thumb points along the torque vector.

Counter-clockwise — thumb out of the page — is **positive**.

Clockwise — thumb into the page — is **negative**.

This is a convention, not a law. It is the usual one, and this course will hold to it, but a paper may choose the other and will say so. Always read the axis labels on someone else's graph before you interpret the sign.



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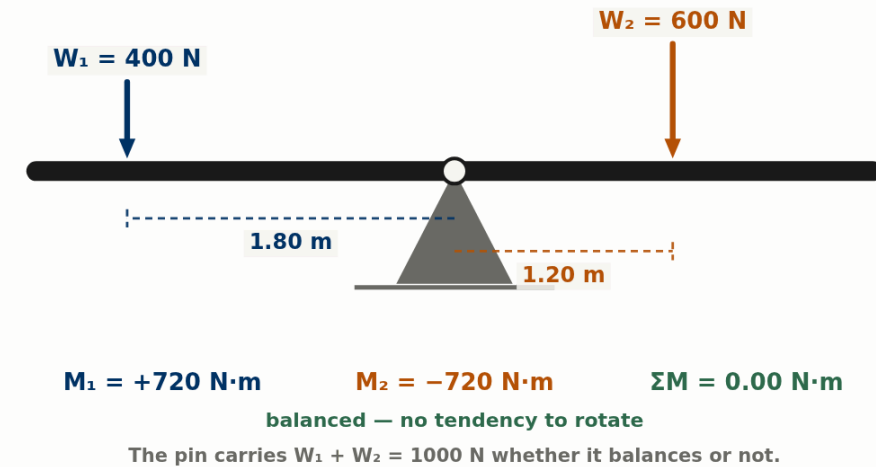
Adding torques

Torques about the **same axis** add, with their signs.

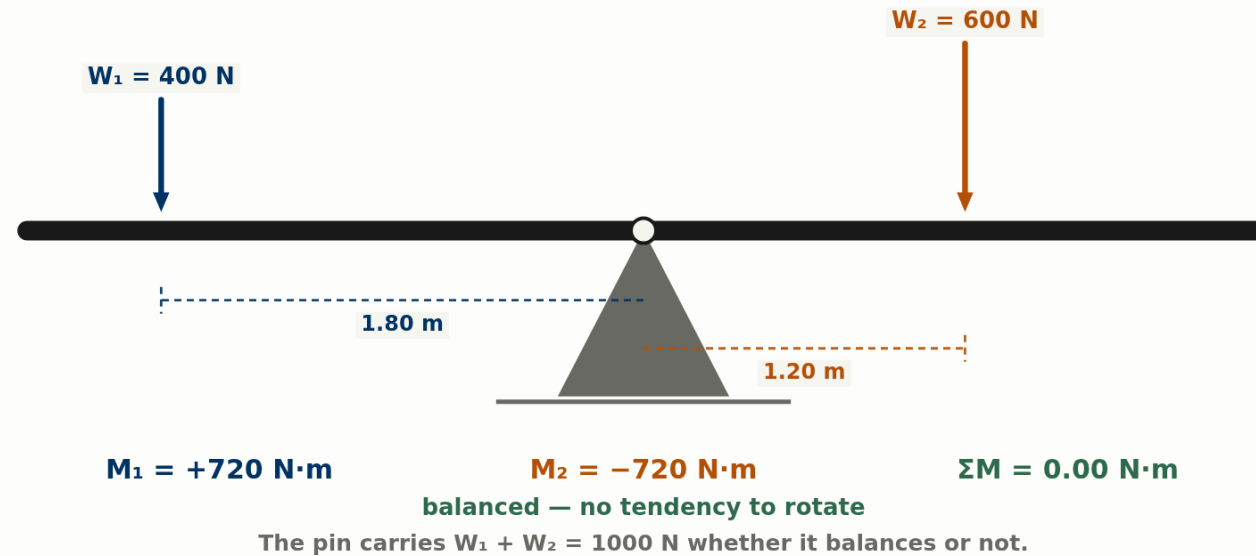
$$\Sigma M = M_1 + M_2 + \dots$$

- Two torques in the same direction reinforce.
- Equal and opposite torques sum to zero and the object does not start to rotate — though both forces are still there, and both are still being resisted by the structure.
- A small force on a long moment arm balances a large force on a short one. That is a lever, and it is the whole of the seesaw.

Only torques about the same axis may be added. Two torques about different axes are two different quantities and adding them is meaningless.



Balancing a lever



W_1 at **1.80 m**
 W_2 weight **600 N**

W_2 at **1.20 m**

W_1 is fixed at **400 N**. Balance is $W_1 d_1 = W_2 d_2$ — a product, so halving a distance exactly doubles the weight it takes. Find a balance with very unequal weights and read the distance ratio: that is the lever's mechanical advantage, and the same number turns up as a muscle's effective mechanical advantage later on.

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Worked example — the moment about O

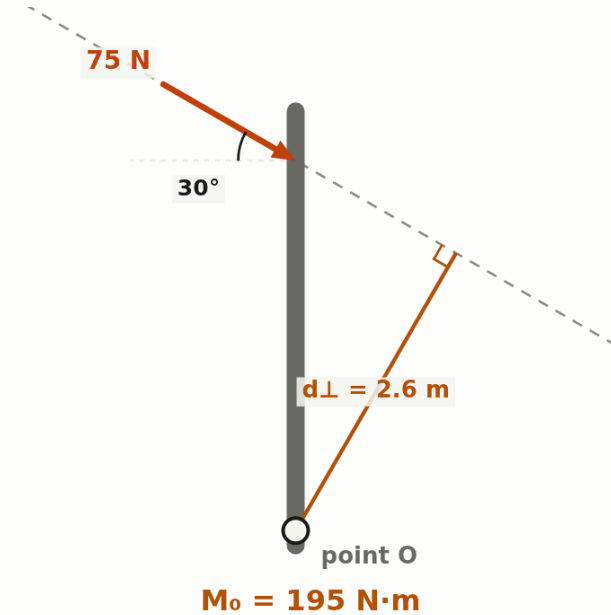
A force of **75 N** acts at the end of a bar **3 m** long, at **30°** to the perpendicular. Find the moment about the axis at O.

Solution A — resolve the distance. Find the perpendicular distance from O to the line of action, then multiply by the whole force.

$$d_{\perp} = (3 \text{ m})(\cos 30^{\circ}) = 2.6 \text{ m}$$

$$M_O = (75 \text{ N})(2.6 \text{ m}) = \mathbf{195 \text{ N}\cdot\text{m}}$$

Note which quantity the cosine is applied to. Here the force is used whole and the *distance* is cut down to the part of it that is perpendicular to the force.



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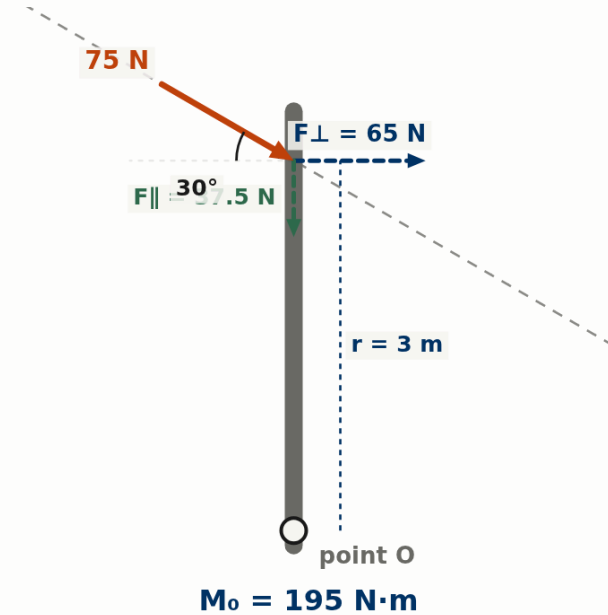
Worked example — the same answer the other way

Solution B — resolve the force. Split the force into a component along the bar and one across it. Only the perpendicular part turns anything; multiply it by the whole distance.

$$F_{\perp} = (75 \text{ N})(\cos 30^{\circ}) = 65 \text{ N}$$

$$M_O = (65 \text{ N})(3 \text{ m}) = \mathbf{195 \text{ N}\cdot\text{m}}$$

Both routes give 195 N·m — use whichever is easier to draw, never both at once. The other component, 37.5 N, runs along the bar through O: no moment arm, no moment, but it is still compressing the bar.



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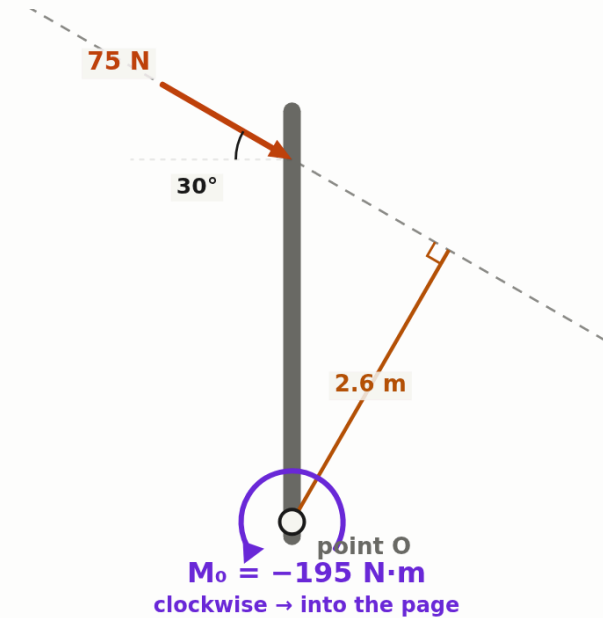
Worked example — and the direction

A magnitude is only half an answer. Apply the right-hand rule: curl your fingers the way the force tends to turn the bar about O.

The force turns the bar **clockwise**. Thumb points **into the page**.

$$M_O = -195 \text{ N}\cdot\text{m}$$

Written as a vector, the moment points along the axis of rotation — into the page — which is at right angles to both the force and the moment arm. Torque vectors never point the way the force points.

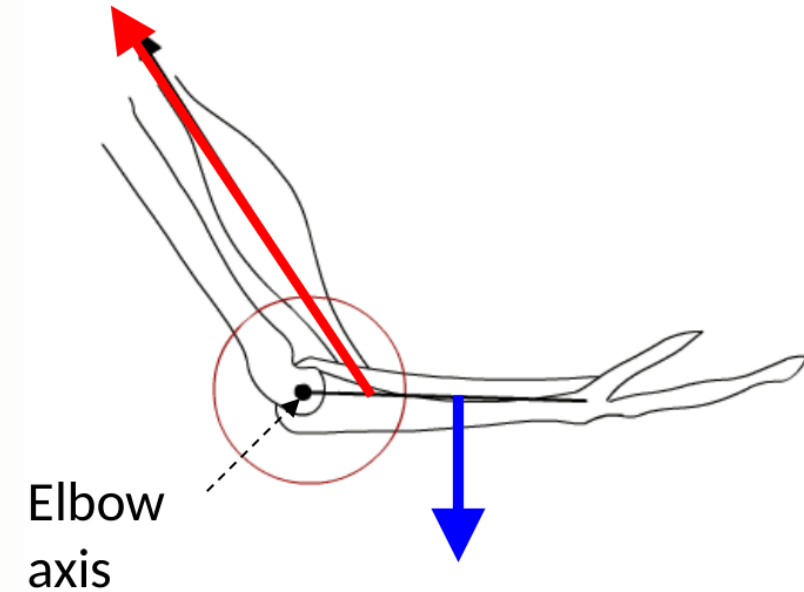


Torque and muscular force

The turning of our limbs is caused by muscular contraction. A muscle attaches a certain distance away from a joint — the axis of rotation — and pulls.

- The **tendon** tells you the line of action. A muscle can only pull along its own line, and the tendon is where you can see that line.
- The **insertion** gives the point of application.
- The moment arm is then the perpendicular from the joint centre to the tendon's line — a few centimetres, usually, and always much shorter than the limb it moves.

Short moment arms are the body's bargain: a muscle that shortens a little moves a hand a long way, and pays for it by having to pull very hard.



The red arrow is the muscle's line of action. The blue arrow is the load. Both make a moment about the same elbow axis, in opposite senses.

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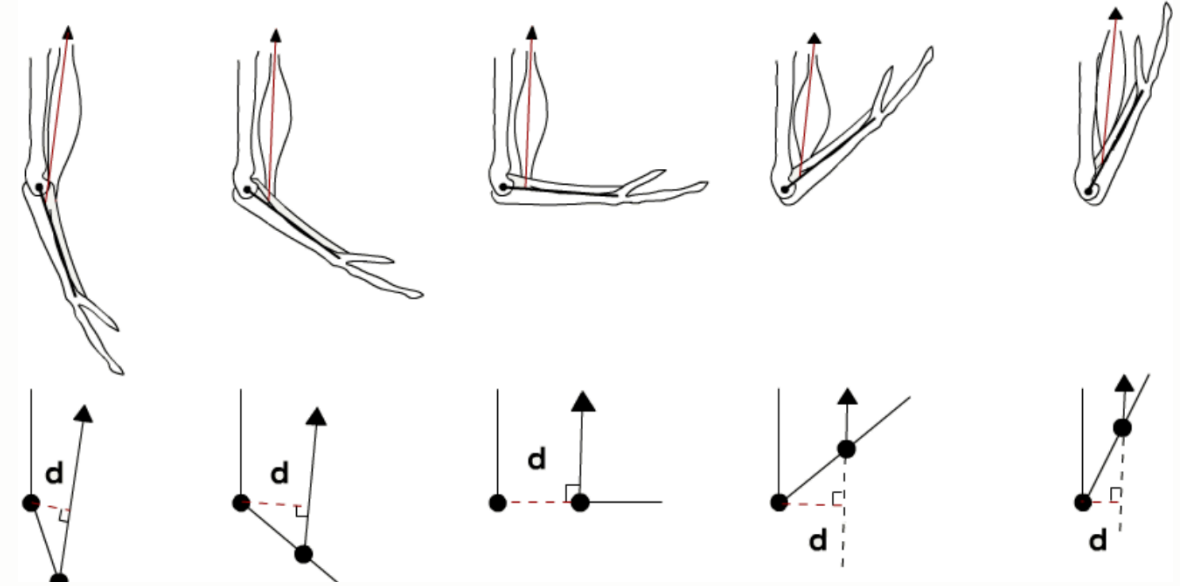
Moving the limb changes the moment arm

Moving a limb changes the **orientation of the muscle's line of action relative to the bone**, and so changes the moment arm.

The same muscle force therefore produces a **different joint torque** at a different joint position. This is why you are strong at some angles and weak at others — and it is a purely geometric effect, before any physiology.

$$d_{\perp} = r \cdot \sin \theta$$

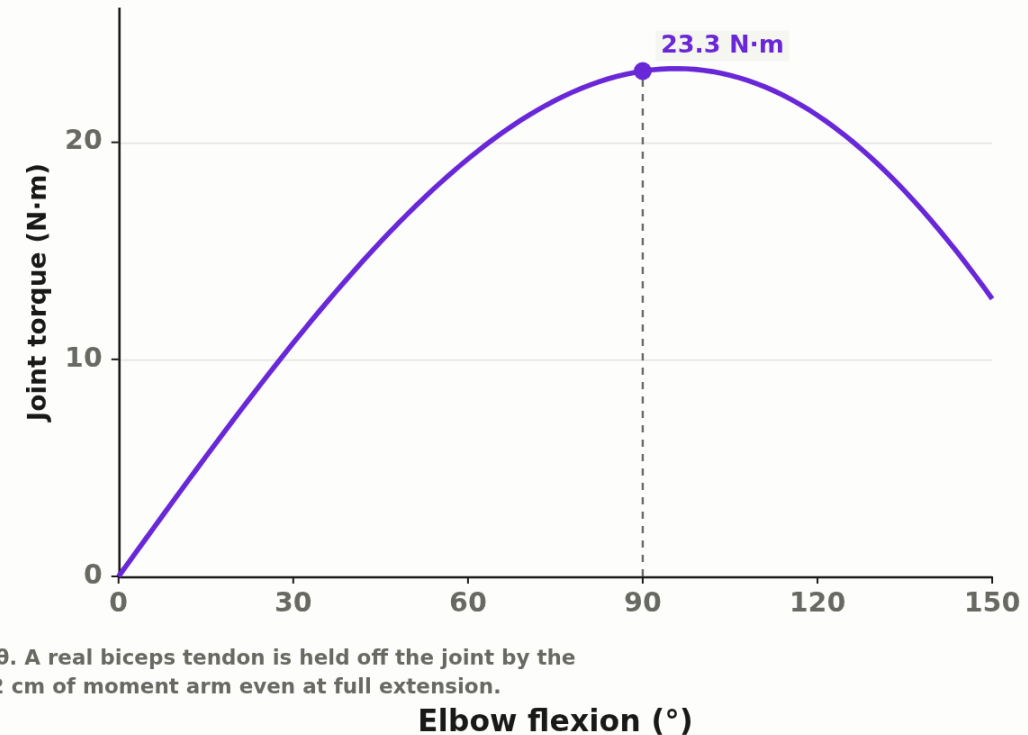
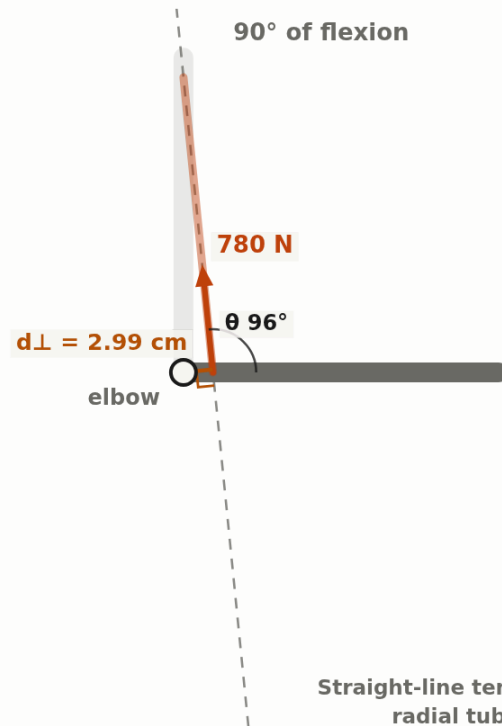
where **r** is the distance from the joint centre to the insertion and **θ** is the angle between the tendon and the bone. At $\theta = 90^\circ$ the moment arm is the whole of **r**; at 0° or 180° it is nothing.



Five positions, one muscle force. The bottom row is the same picture with everything but the geometry taken away.

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The biceps through the range



Straight-line tendon model: $d_\perp = r \sin \theta$. A real biceps tendon is held off the joint by the radial tuberosity and keeps 1.5-2 cm of moment arm even at full extension.

Elbow flexion 90°
Load in the hand none

Muscle force 780 N

With no load the right-hand panel is the **torque this muscle force can make** at each joint angle — the same 780 N doing very different work. Add a load and the panel switches to the **muscle force the load demands**: two moment arms are now changing at once, and their ratio is the effective mechanical advantage.

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Example problem — the biceps at the elbow

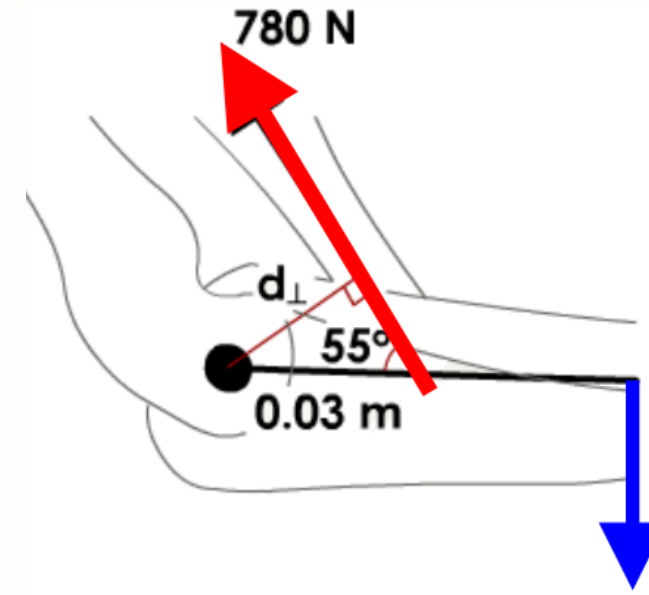
The biceps brachii develops a force of **780 N**. It inserts **0.03 m** from the elbow joint axis, and the tendon lies at **55°** to the forearm.

Find the torque about the elbow.

Start by drawing a free body diagram.

Mark the axis, the point of application, the force with its direction, and the angle. Extend the line of action as a dashed line well past the insertion — you will need it to find the perpendicular.

Do not reach for a formula until the diagram is drawn. Nearly every error in this kind of problem is a geometry error that a careful sketch would have caught.



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Step 1 — the moment arm

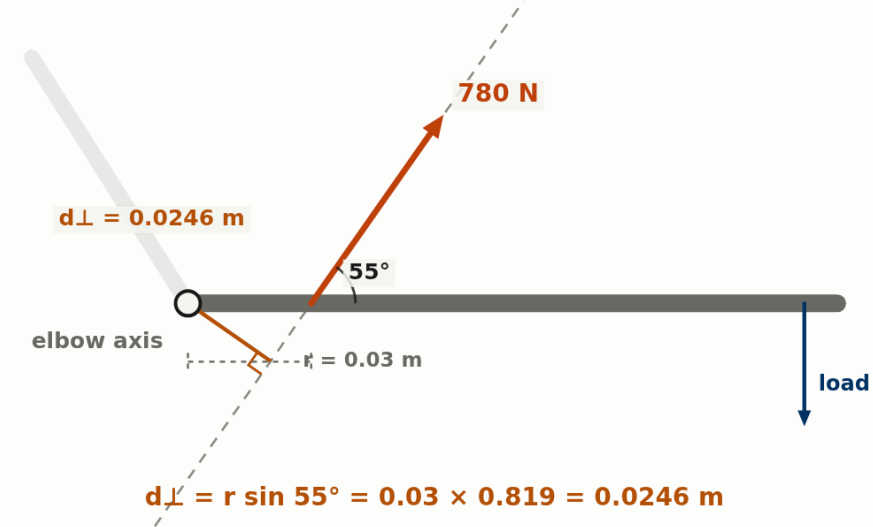
The moment arm is the perpendicular distance from the elbow axis to the line of action of the muscle force.

$$d_{\perp} = r \cdot \sin \theta$$

$$d_{\perp} = (0.03 \text{ m})(\sin 55^{\circ})$$

$$d_{\perp} = \mathbf{0.0246 \text{ m}}$$

Just under two and a half centimetres — noticeably less than the three centimetres the tendon actually inserts at, because the tendon is pulling at an angle rather than squarely across the bone.



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Step 2 — the torque

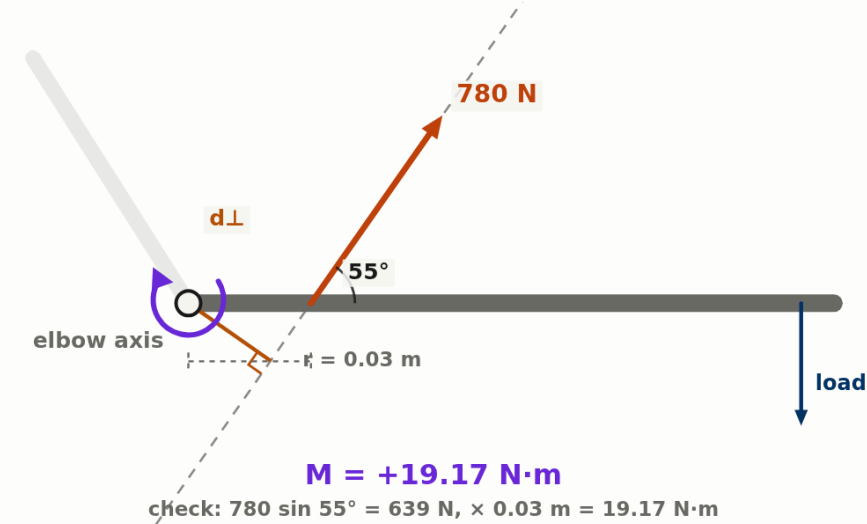
$$M = F \cdot d_{\perp}$$

$$M = (780 \text{ N})(0.0246 \text{ m})$$

$$M = \mathbf{+19.17 \text{ N}\cdot\text{m}}$$

The biceps flexes the elbow, turning the forearm counter-clockwise as drawn, so by the right-hand rule the torque vector points out of the page and the sign is **positive**.

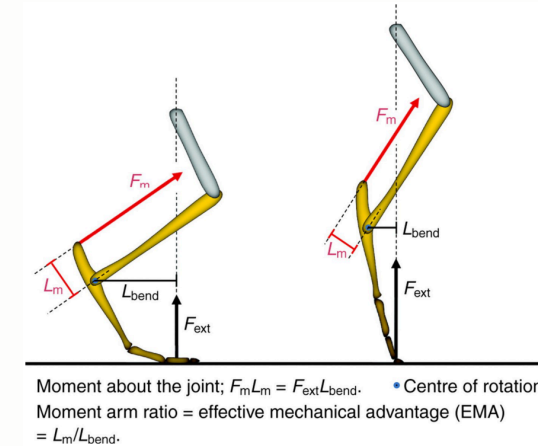
Check it the other way. $F_{\perp} = 780 \sin 55^{\circ} = 639 \text{ N}$, and $639 \text{ N} \times 0.03 \text{ m} = 19.17 \text{ N}\cdot\text{m}$.
Same answer, as it must be.



Why we go to this trouble

In biomechanics we quantify forces and torques in order to answer questions we cannot answer by looking:

- **How will the joint move?** The net moment at a joint, not any single muscle, determines the angular acceleration.
- **How will the body translate and rotate?** Linear kinetics gave the first half of that; this is the other half.
- **What is happening to the tissues?** Bones, cartilage, tendons and ligaments all fail at loads we can estimate but cannot see.
- **How hard is a muscle working?** We cannot measure muscle force in a living person without surgery. We can estimate it, and a moment arm is how.



Moment about the joint: $F_m L_m = F_{ext} L_{bend}$. The ratio of the two moment arms is the **effective mechanical advantage**.

Inverse dynamics

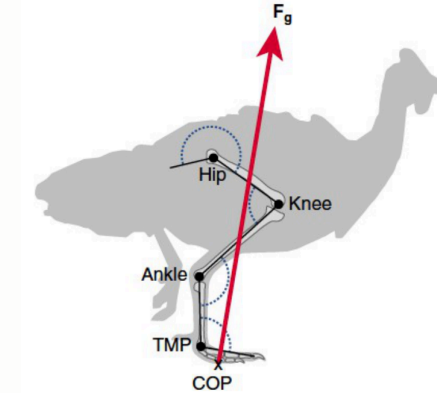
We cannot measure the force in a muscle, or the moment at a joint, directly. We can measure the things at the outside of the system — where the segments are, and what the ground is doing — and work inwards.

That is **inverse dynamics**: from the measured motion and the measured external forces, compute the net moment that must have been acting at each joint.

- Start at the ground with the force plate, and the foot.
- Solve the foot, and pass the result up to the ankle.
- Solve the shank, and pass it up to the knee. Then the thigh, and the hip.

It gives the **net** moment — one number per joint, the sum of every muscle, ligament and contact force there. It cannot tell you which muscle made it, or whether an antagonist was active at the same time.

M. A. Daley, G. Felix and A. A. Biewener



The measurement, on an animal: joint centres from markers, the force vector from a plate, and everything else computed. Daley, Felix & Biewener.

External moments and internal moments

Two moments at every joint, and they are not the same thing.

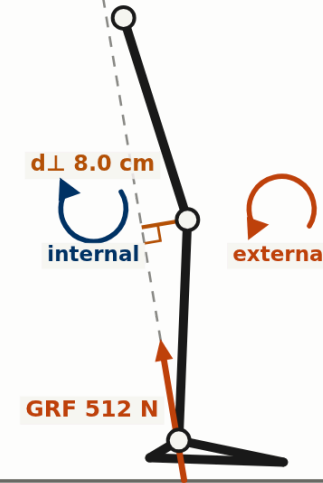
External moment — made from outside: gravity and the ground reaction force. What the world is doing to the joint.

Internal moment — made from inside: muscles, and at the end of range ligaments and capsule. The body's answer.

When the joint is not accelerating, these are **equal and opposite**. An external flexor moment is met by an internal extensor moment.

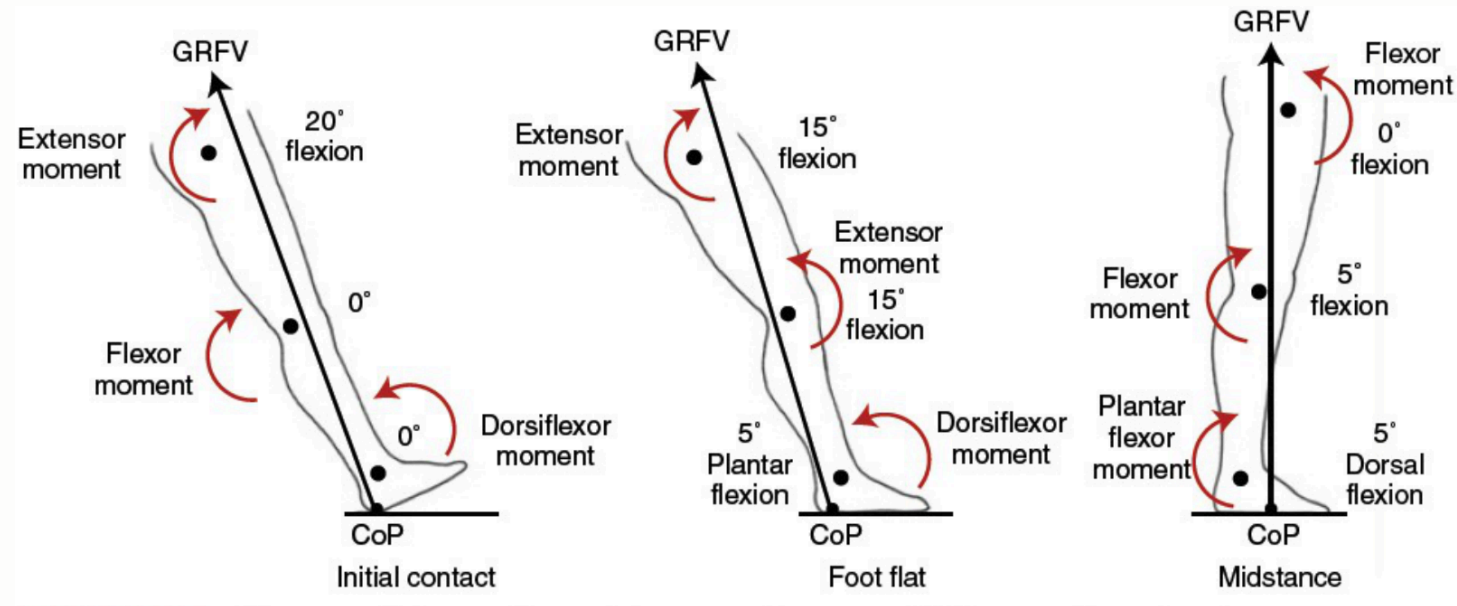
Gait graphs almost always plot the **internal** moment, because that is the one that names a muscle group. A curve labelled "extensor" means the extensors were working — so the external moment was trying to flex.

Loading response, Subject16 — measured



the line passes 8.0 cm behind the knee, so the ground makes an **EXTERNAL FLEXOR moment** and the quadriceps answer with an **INTERNAL EXTENSOR moment of 0.74 N·m/kg**

Which side of the joint does the line pass?



- At **initial contact** the vector passes *behind* the ankle, so the external moment tends to plantarflex it and the dorsiflexors answer.
- In **early stance** it swings *behind* the knee — tending to flex it — and the quadriceps answer. It moves ahead of the knee again by late stance.
- At the same instant it passes *in front of* the knee and hip, tending to extend them.
- From about **midstance** it passes *behind* the hip and stays there until swing begins.

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A stick figure is shown in a lunge position. A vertical dashed line represents the center of mass (CoM). A red dot on the ground represents the center of pressure (CoP). A red arrow points from the CoP to the CoM. The distance between them is labeled $d_{\perp} = 6.6 \text{ cm}$. The weight of the figure is labeled 580 N (1.07 BW) .



Gait cycle 15%

Inverse dynamics

23

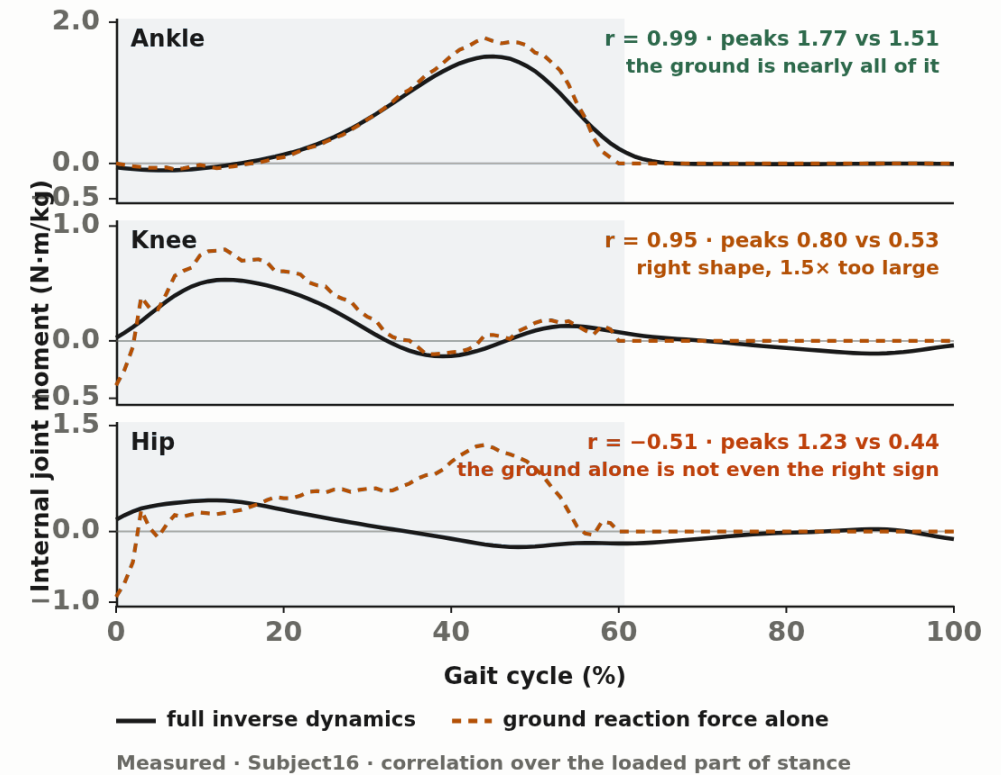
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How much is the ground, and how much is the limb?

If you know the ground reaction force and where each joint is, you can compute the moment the ground alone makes about that joint. Is that the joint moment?

Compare the ground-alone estimate with the full inverse-dynamics result, on the same measured stride:

- **Ankle** — nearly the whole story. The two agree to within about ten per cent through stance. The foot is light and barely accelerates, so there is almost nothing else to account for.
- **Knee** — the right shape, the wrong size. The ground alone overestimates the peak by roughly half, because the shank and foot have weight of their own and are being swung.
- **Hip** — wrong. Not merely too large: for much of stance it has the wrong *sign*. Everything below the hip has to be carried and accelerated, and that dominates.



Subject16

Subject15

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Reading a moment–time graph

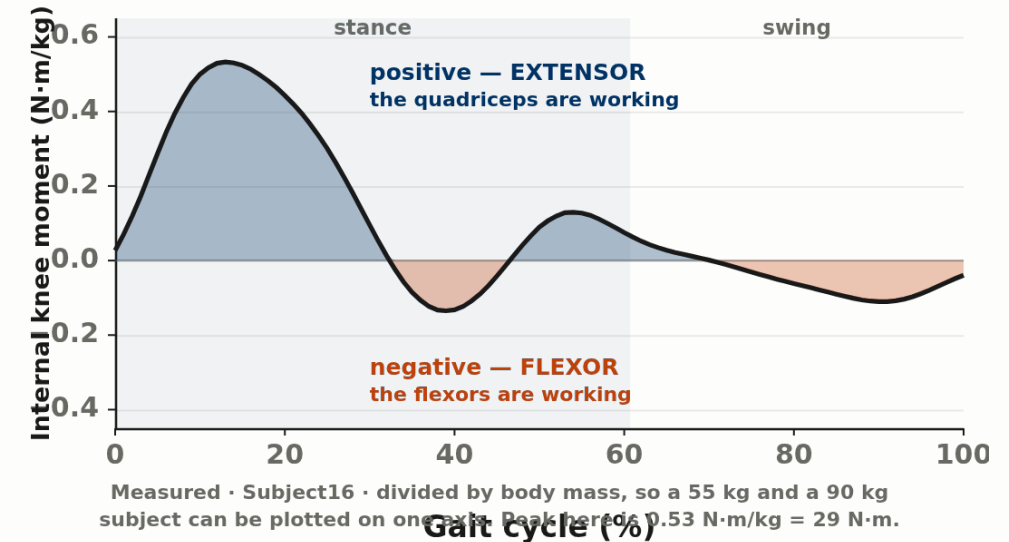
A moment–time graph is read in two steps: **which side of zero**, and **how far from it**.

The sign names a direction of rotation, and at a joint that means a muscle group — which one depends on the joint and the axis:

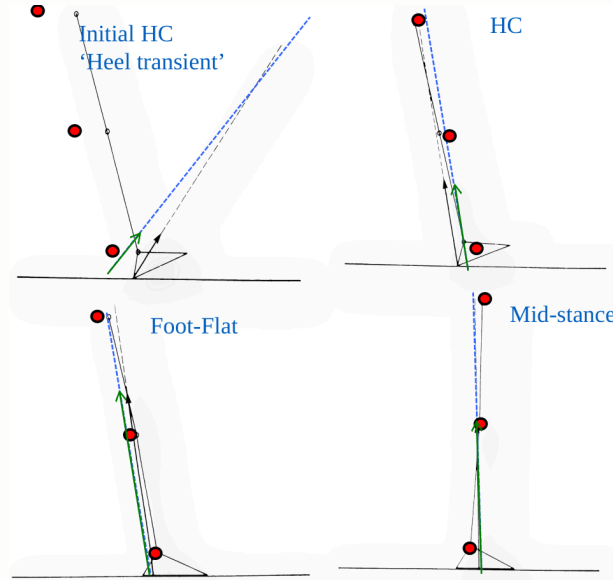
- **Flexion / extension** — the sagittal plane, and nearly everything in this lecture.
- **Inversion / eversion** — the frontal plane at the foot.
- **Internal / external rotation** — the transverse plane.

Every joint has a moment about all three axes at once. We plot the sagittal one because in walking it is much the largest, not because the others are zero.

Before reading any moment graph, find out three things: internal or external, which way is positive, and whether it is per kilogram. All three change the number on the axis.



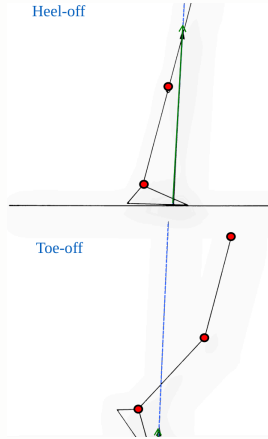
Gait kinetics — contact to midstance



To account for a joint moment you need **both** the external force and what the muscles are doing about it.

- **Initial heel contact** — the "heel transient", a very brief spike as the heel decelerates. It is largely passive, over in a few milliseconds, and the limb has no time to respond to it.
- **Heel contact** — the vector is steeply inclined and passes well in front of the hip. The hip extensors are already working.
- **Foot flat** — the vector has swung behind the knee. Quadriceps.
- **Midstance** — the vector is nearly vertical and passes close to the knee, so there is little to do there; meanwhile the centre of pressure has moved forward of the ankle and the plantarflexors begin to build.

Gait kinetics — heel-off to toe-off

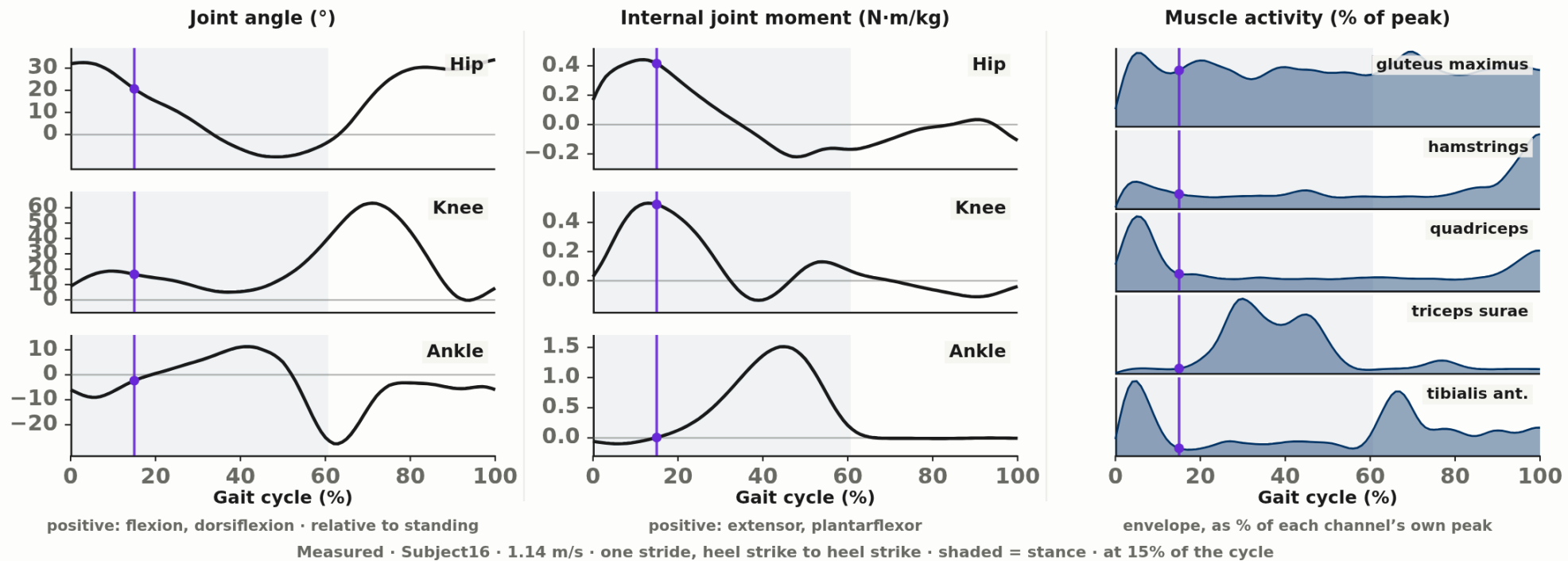


- **Heel-off** — the centre of pressure is now well forward under the forefoot while the ankle is still behind it. The moment arm at the ankle is at its largest and the vertical force is near its second peak at the same time. This is where the plantarflexor moment peaks, and it is by far the largest moment anywhere in the leg.
- **Toe-off** — the limb is unloading. The force falls away to nothing over the last few per cent of stance, and with it every moment the ground was making.

Two things multiply to make the ankle peak: a **large force** and a **long moment arm**, arriving together. Either one alone would be unremarkable.

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Angles, moments and muscles, one stride



Gait cycle — 15%

Subject16

Subject15

▶ Walk

Read one instant across all three columns. At **10–15%** the knee is flexing, the knee moment is extensor and the quadriceps are on — the ground is folding the knee and the quadriceps are controlling the fold, lengthening while they resist. Two honest gaps: the **gluteus maximus** channel barely modulates in either subject, which is a poor recording and not a quiet muscle, and **iliopsoas** — which the textbook panel shows driving the hip flexor moment in pre-swing — is too deep for any surface electrode to reach.

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What the three panels are for

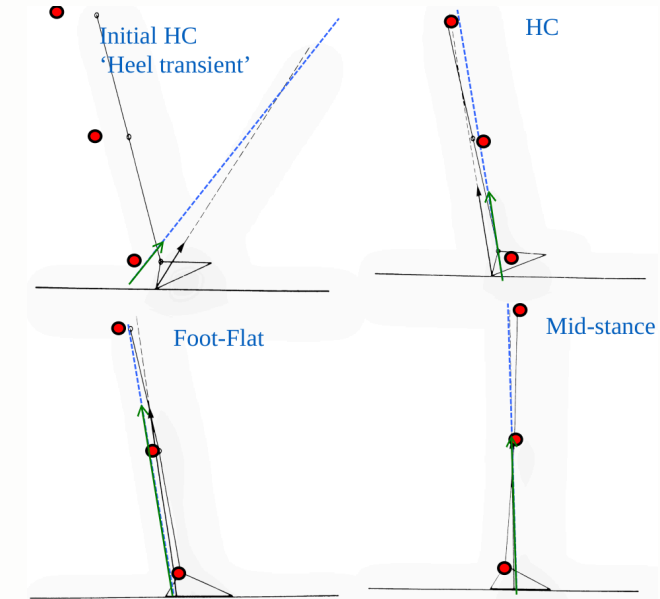
A joint moment on its own is ambiguous. Read with the angle and with the muscle activity, it is not.

Moment and movement in the same direction — shortening. Concentric, doing positive work on the body.

Moment and movement in opposite directions — lengthening under load. Eccentric, absorbing energy.

Most of what the leg does in stance is the second kind: the quadriceps at loading response and the plantarflexors through midstance both lengthen while resisting, and only the plantarflexors turn concentric at the end.

The product of moment and angular velocity is **joint power** — positive for generation, negative for absorption. That is the next lecture.

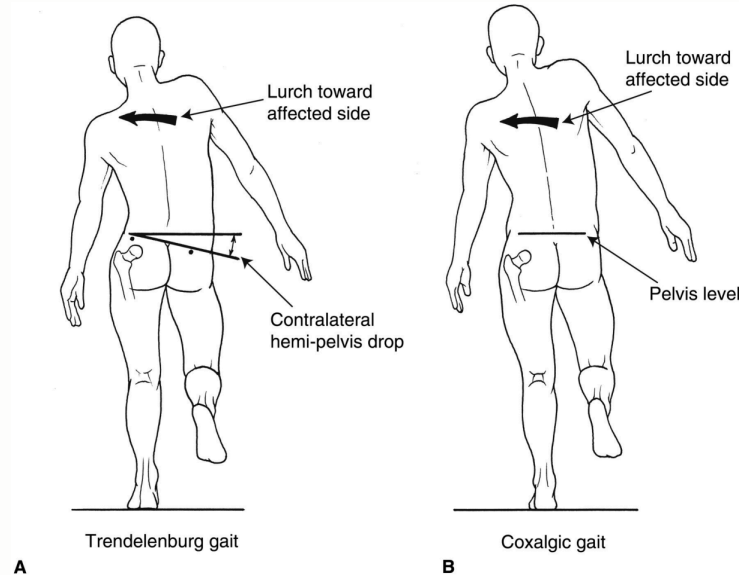


The same four instants. Which side of the joint the line falls on gives the sign; the angle panel tells you which way the joint was going at the time.

Common gait abnormalities

Three ways of moving a line of action, and what each one buys

Lateral trunk bending



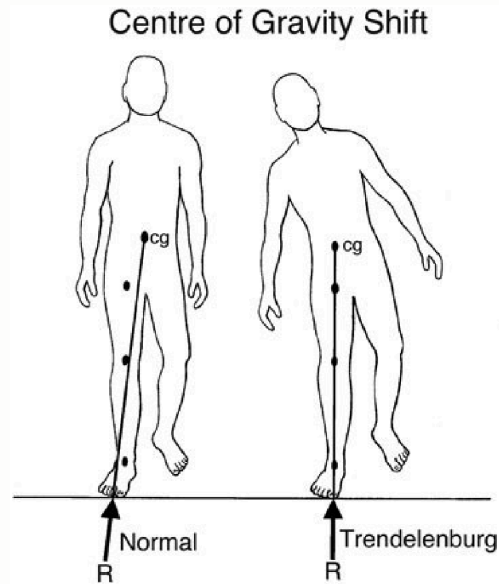
Bending the trunk **towards the side of the supporting limb** during stance is lateral trunk bending, or ipsilateral lean.

Most commonly seen in:

- **Trendelenburg gait** — the abductors on the stance side cannot hold the pelvis, so the opposite hemi-pelvis drops *and* the trunk lurches to compensate.
- **Coxalgic gait** — "painful hip". The pelvis stays level; the lean is there to reduce the load, not because the pelvis cannot be held.

The lean is the same in both. What distinguishes them is whether the pelvis drops with it — which is why you watch the pelvis, not the shoulders.

Shifting the centre of gravity



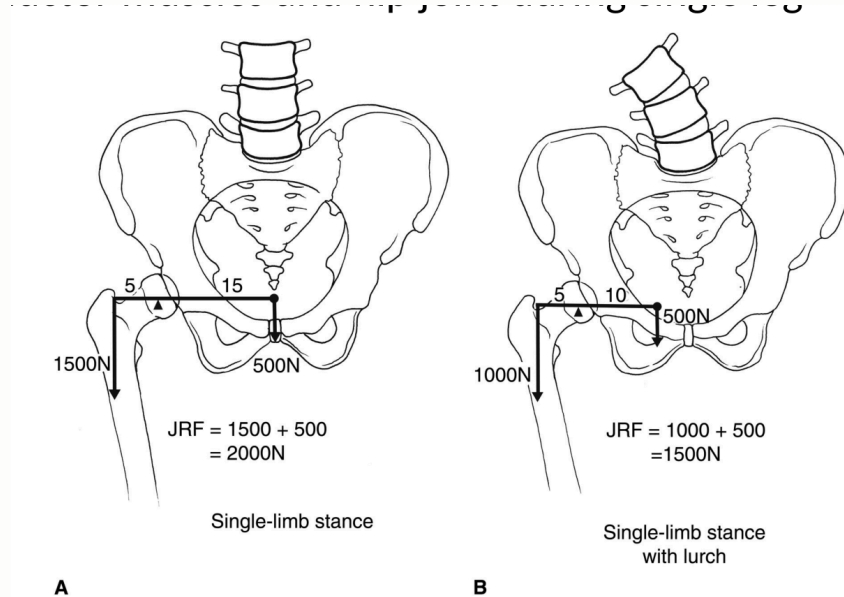
Shifting the body's centre of gravity more **vertically over the hip joint** reduces the torque that the ground reaction force makes about that hip.

Nothing about the force has changed. The person still weighs the same, and the ground still pushes up with the same magnitude. What has changed is the **moment arm**: the perpendicular distance from that line to the hip.

$$M = F \cdot d_{\perp} \rightarrow \text{halve } d_{\perp}, \text{ halve } M$$

And because the abductors must match that moment on a very short lever of their own, a small change in $d_{\perp}$ makes a large change in muscle force.

The arithmetic of the lurch



Single-limb stance. Body weight above the hip acts 15 units from the joint; the abductors act 5 units the other side.

$$F_{ab} \times 5 = 500 \text{ N} \times 15$$

$$F_{ab} = 1500 \text{ N}$$

$$\text{JRF} = 1500 + 500 = \mathbf{2000 \text{ N}}$$

With the lurch, the body weight arm falls to 10 units.

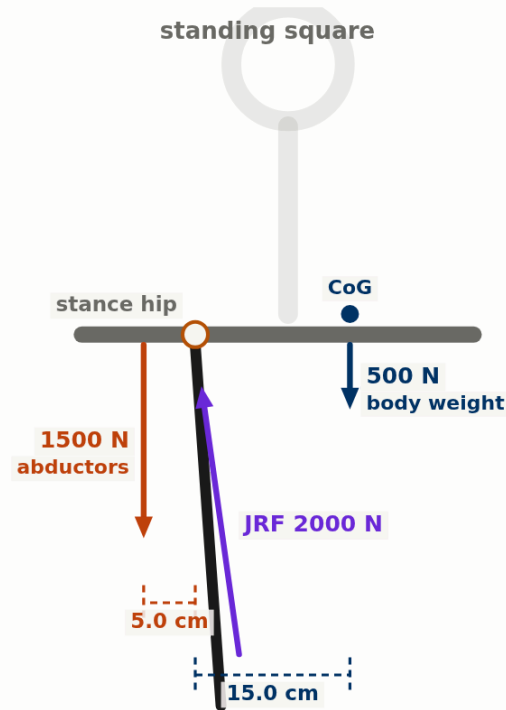
$$F_{ab} \times 5 = 500 \text{ N} \times 10 \rightarrow 1000 \text{ N}$$

$$\text{JRF} = 1000 + 500 = \mathbf{1500 \text{ N}}$$

Common causes: painful hip (osteoarthritis, rheumatoid arthritis), hip abductor weakness, leg-length discrepancy, abnormal hip joint.

Angular Kinetics 1

What a lean is worth



Moments about the stance hip

$$F_{ab} \times 5.0 \text{ cm} = 500 \text{ N} \times 15.0 \text{ cm}$$

$$F_{ab} = 1500 \text{ N} \text{ (2.50 body weights)}$$

$$JRF = F_{ab} + W = 2000 \text{ N}$$

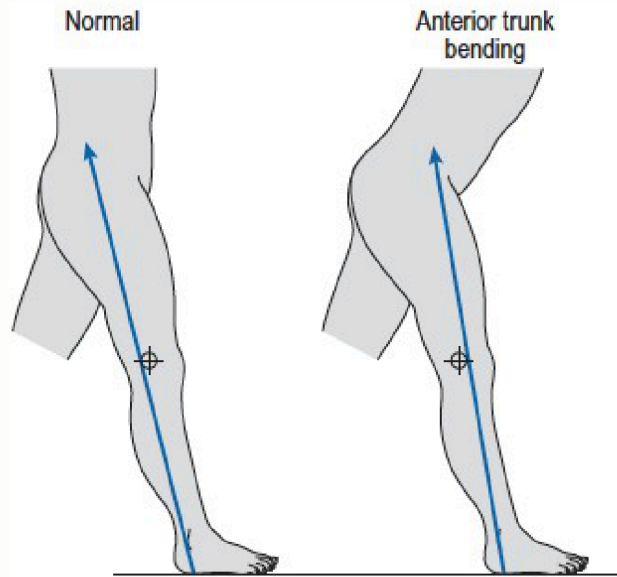
the femoral head carries 3.33 body weights

Standing square with a 5 cm abductor lever: his figure, 1500 N and 2000 N. Lean the trunk, or shorten the lever, and watch both numbers move.

Trunk lean 0.0° Abductor moment arm 5.0 cm

A lean of eight or nine degrees — barely visible across a room — takes about a third off the hip load. Shortening the **abductor moment arm** sends the required force up steeply, which is why restoring the offset matters in a hip replacement. And a cane in the **opposite** hand, carrying only 15% of body weight on a very long lever, beats the lurch; in the **same** hand it makes everything worse.

Anterior trunk bending



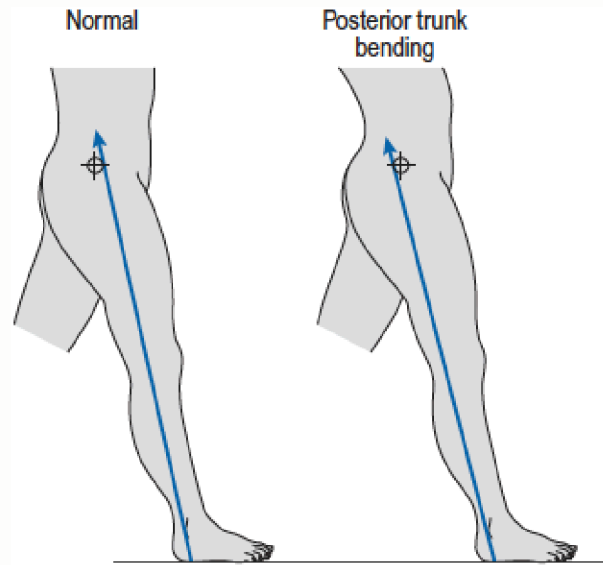
The subject flexes the trunk forwards early in stance.

Normal. In early stance the line of action of the ground reaction force passes *behind* the knee axis. The external moment tends to flex the knee, and the quadriceps oppose it with an internal extensor moment.

If the quadriceps are weak or paralysed, that external flexor moment would collapse the knee. Leaning the trunk forward moves the body's centre of gravity ahead of the knee, carrying the line of action with it.

Now the ground is *extending* the knee rather than flexing it. The joint is held by its own posterior structures, and the quadriceps are not needed at all.

Posterior trunk bending



The subject extends the trunk backwards early in stance.

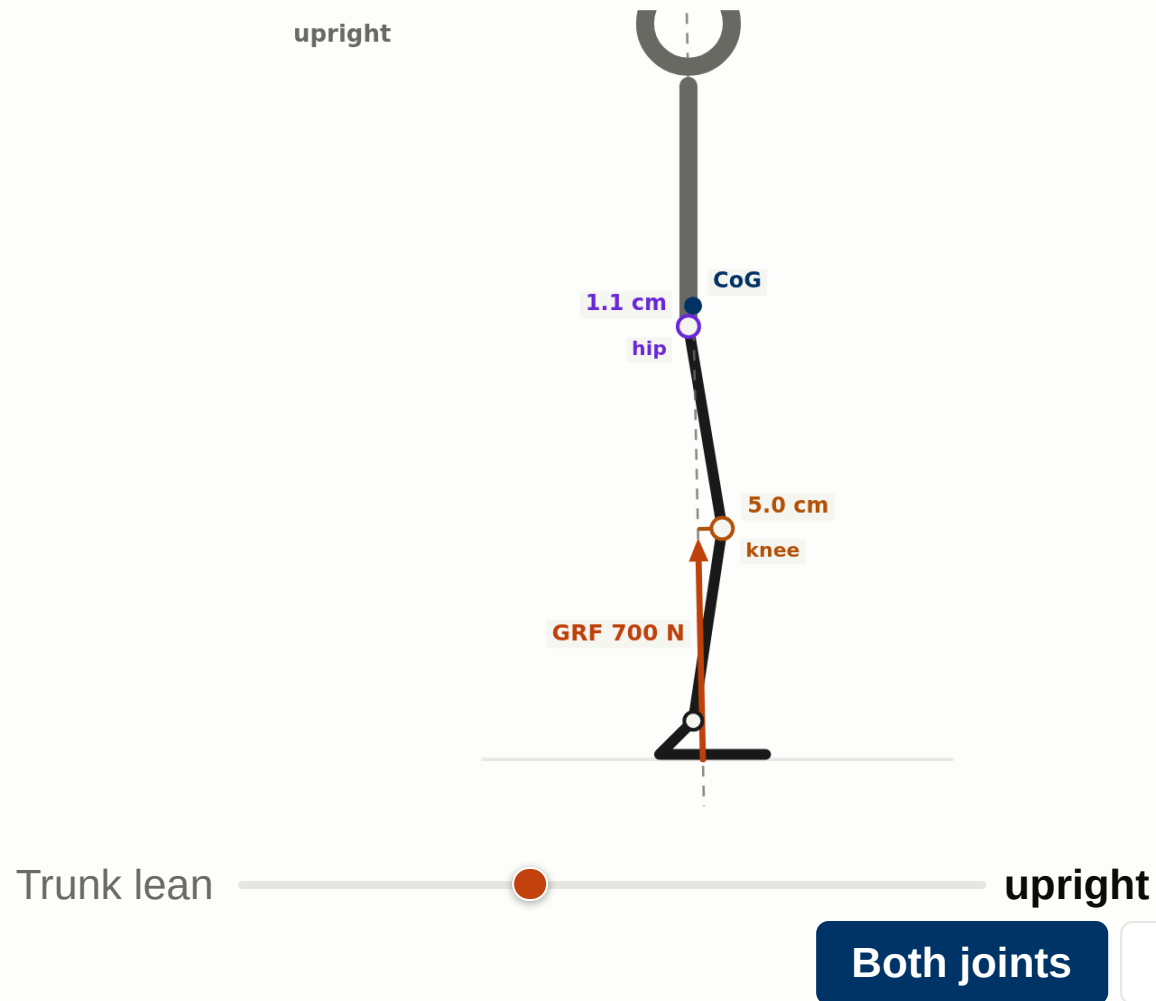
Normal. In early stance the line of action passes *in front of* the hip axis, so the external moment tends to flex the hip and the hip extensors — gluteus maximus and the hamstrings — oppose it.

If gluteus maximus is weak or paralysed, leaning the trunk backwards brings the line of action *behind* the hip axis.

The external moment is now an extensor moment, and the hip extensors have nothing to do. Sometimes called a gluteus maximus lurch.

Angular Kinetics 1

Moving the line of action



Knee

the line passes 5.0 cm behind it
external flexor moment of 35 N·m
the quadriceps must work to resist it

Hip

the line passes 1.1 cm in front of it
external flexor moment of 7 N·m
the hip extensors must work to resist it

Static model: in quiet stance the resultant must pass through the centre of gravity.
True only while nothing is accelerating — close enough in early stance to make the point.

Lean **forward** and the line swings ahead of the knee: the external moment reverses and the quadriceps are unloaded, which is what a patient with a weak quadriceps finds without being taught. Lean **back** and the same happens at the hip. The two cannot be satisfied at once — whichever way you lean, one of the two muscle groups has more to do.

What to take away

- **Torque is force times perpendicular distance.** Both words matter — perpendicular, and to the *line of action*, not to the point of application.
- **Either route gives the same answer.** Resolve the distance or resolve the force. Never both at once.
- **Name the axis first.** The same force has a different moment, and often a different sign, about a different axis.
- **Joint angle changes the moment arm.** The same muscle force makes a different joint torque at a different position, before any physiology is involved.
- **External and internal moments are opposite sides of one balance.** Gait graphs plot the internal one, so a curve labelled extensor means the ground was trying to flex.
- **Which side of the joint the line passes decides the sign** — everywhere. It decides the *size* reliably only at the ankle; at the hip, the limb's own weight and acceleration dominate, which is why inverse dynamics exists.
- **Compensations move a line of action.** Lateral, anterior and posterior trunk bending are one manoeuvre about three different axes, and each unloads a particular muscle at a price paid somewhere else.

BIOMECHANICS (EPHE 341)

Angular Kinetics 1

Next up: angular kinetics 2 — moment of inertia, angular momentum and joint power ·
Measured data: Lencioni et al., level walking